

Report a2q2.pdf

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- Explain how $r(\text{txt}[j \dots j+m-1])$ was updated from previous $r(\text{txt}[j-1 \dots j+m-2])$.

Assuming r is calculated correctly after calculating the r value of the starting window of the text.

Let the old window end at $j+m-2$, and the new start at J .

$Z_{\text{old}} = r(\text{txt}[j-1 \dots j+m-2])$, the algorithm can 'update' the value to find the next

$Z_{\text{new}} = r(\text{txt}[j \dots j+m-1])$ through manipulating Z_{Old} .

The first realization for this is that we can use all 'intermediate' calculated values of the integer mapping, and only need to remove the character leaving the window. That is the char at $\text{txt}[j-1]$.

This is done by subtracting the term: $Z_{\text{Old}} - \text{index}(\text{char}) * \text{Beta}^{m-1}$

Then we 'shift' the characters left, that is multiplying them by Beta to effectively

$$= \left(\sum_{i=1}^k \text{index}(s_i) \times \beta^{k-i} \right)$$

increment k noted in the $R(S)$ definition;

Raising the power of each term by 1.

From this we add the new character entering the window: $\text{Beta}^0 * \text{index}(\text{txt}[j+m-1])$

Effectively this relationship is defined as:

$$Z_{\text{new}} = (Z_{\text{Old}} - \text{index}(\text{txt}[j-1]) * \text{Beta}^{m-1}) * \text{Beta} + \text{index}(\text{txt}[j+m-1])$$

Further we apply the modulo p for each operation as follows:

$$[(r_{\text{old}} - (\text{index}(\text{txt}[j-1]) * (\text{Beta}^{m-1} \bmod p))) * \text{Beta} + \text{index}(\text{txt}[j])] \bmod p$$